



Saffron and crocin ameliorate prenatal valproic acid-induced autistic-like behaviors and brain oxidative stress in the male offspring rats

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Abstract

Autism is a neurobehavioral disease that induces cognitive and behavioral alterations, usually accompanied by oxidative stress in the brain. *Crocus sativus* (saffron) and its active ingredient, crocin, have potent antioxidative effects that may benefit autistic behaviors. This study aimed to determine the effects of saffron extract and crocin against brain oxidative stress and behavioral, motor, and cognitive deficits in an animal model of autism in male offspring rats. 14 female rats were randomly divided into the saline and valproic acid (VPA) groups. Then, they were placed with mature male rats to mate and produce offspring. VPA (500 mg/kg, i.p.) was injected on day 12.5 of pregnancy (gestational day, GD 12.5) to induce an experimental model of autism. 48 male pups were left undisturbed for 29 days. First-round behavioral tests (before treatments) were performed on 30–33 post-natal days (PND), followed by 28 days of treatment (PND 34–61) with saffron (30 mg/kg, IP), crocin (15 or 30 mg/kg, i.p.), or saline (2 ml/kg, i.p.). The second round of behavioral tests (after treatments) was performed on PND 62–65 to assess the effects of the treatments on behavioral and cognitive features. In the end, animals were sacrificed under deep anesthesia, and their brains were dissected to evaluate the brain oxidative stress parameters, including malondialdehyde (MDA), glutathione (GSH), and catalase (CAT). VPA injection into female rats increased anxiety-like behaviors, enhanced pain threshold, impaired motor functions, disturbed balance power, increased MDA, and decreased GSH and CAT in their male offspring. 28 days of treatment with saffron or crocin significantly ameliorated behavioral abnormalities, reduced MDA, and increased GSH and CAT levels. Brain oxidative stress has been implicated in the pathophysiology of autistic-like behaviors. Saffron and crocin ameliorate anxiety-like behaviors, pain responses, motor functions, and brain oxidative stress parameters in an experimental model of autism. Saffron and crocin may hold promise as herbal-based pharmacological treatments for individuals with autism. However, further histological evidence is needed to confirm their efficacy.

Keywords Autism · Saffron · Crocin · Valproic acid · Oxidative stress · Cognition · Anxiety

Introduction

Autism is a neurodevelopmental disorder characterized by various behavioral symptoms, including impairments in social and verbal performances and limited interests and activities associated with repetitive and stereotyped patterns (Landa 2008). Previous studies have shown that genetic factors, immune system disorders (Chaste and Leboyer 2012), maternal infectious diseases (Patterson 2011a, b), and the level of lipid peroxidation, which is a prominent marker of oxidative stress (Chauhan and Chauhan 2006), are involved in the pathophysiology of autism. Autism is associated with increased motor activity, repetitive movements, stress, abnormal social behaviors,

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irritability, and reduced pain thresholds (Lord et al. 2018; Schneider et al. 2008). Valproic acid (VPA) administration is a standard animal model for inducing autism that has been used extensively over recent years (Nicolini and Fahnestock 2018). This model is similar to autism in humans regarding impairment in the cranial nerve motor nuclei in the brainstem and decreased Purkinje cells in the cerebellum (Martin et al. 2010). On the other hand, children exposed to VPA during pregnancy are eight times more susceptible to autism (Chaste and Leboyer 2012). VPA acts as an antiepileptic drug by acting on the GABA receptors and sodium channels, increasing apoptosis and impairing neurodevelopment, leading to the increased density of glutamatergic neurons (Fukuchi et al. 2009; Kim et al. 2016, 2014) and decreased cell proliferation in some regions of the brain (Kataoka et al. 2013; Phiel et al. 2001). In this disease, serotonergic neuronal maturation is impaired and may lead to changes in serotonin levels in the brain (Miyazaki et al. 2005; Narita et al. 2002). Although medication and behavioral therapy are common treatments for autism, no effective treatment has been identified yet (Aali et al. 2014). Since this disorder is associated with oxidative damage to the brain, researchers have recently focused on using antioxidants, especially those present in medicinal plants (Kim et al. 2013a, b).

Saffron, derived from the flower of *Crocus sativus*, also known as saffron crocus, is mainly cultured in Iran, India, Tibet, Mexico, Spain, and Australia (Hossein-zadeh et al. 2008; Pitsikas and Sakellariadis 2006; Pitsikas et al. 2007). It combines a bitter taste, aromatic smell, and red color. The main ingredients of its extract include a variety of carotenoids, such as crocetin, safranal, picrocrocin, and crocin (Schmidt et al. 2007). Saffron extract and crocin improve learning and memory deficits induced by chronic stress or ethanol and prevent scopolamine-dependent memory loss and cognitive impairments (Hossein-zadeh et al. 2008; Pitsikas and Sakellariadis 2006; Pitsikas et al. 2007). Intraperitoneal administration of crocin improves serum antioxidant markers in rats (Tamaddonfard et al. 2013). Crocin potentiates antioxidative systems and increases antioxidative activities, including glutathione peroxidase, glutathione reductase, and superoxide dismutase leading to decreased oxidative injuries and lipid peroxidation (Adabizadeh et al. 2019; Yaribeygi et al. 2018a, b, c; Yaribeygi et al. 2019). Since these antioxidant enzymes are usually reduced in autism (Mandic-Maravic et al. 2019), saffron and crocin may preserve antioxidant capacity and prevent oxidative damage associated with behavioral impairments in autism. So, the present study aimed to determine the effects of saffron and its active ingredient, crocin, on VPA-induced brain oxidative stress damage and behavioral deficits in an animal model of autism.

Materials and methods

Animals

This study was conducted using 14 female Wistar rats as mothers and 48 male pups as their offspring. Pups were kept (5 per cage) in a 12/12 h light–dark cycle, at 22 to 24 °C, with free access to water and food. The Ethical Review Board of Semnan University of Medical Sciences, Semnan, Iran, approved this study (IR.SEMUMS.REC.1397.144). All experiments were conducted in accordance with the National Institutes of Health Guide for the Care and Use of Laboratory Animals.

Induction of autistic-like features

Female rats (mothers; $n = 14$, weight 220–240 g) were randomly divided into two groups, including SAL (to produce healthy offspring) and VPA (to produce autistic offspring). Then, the female rats were randomly placed with mature male rats to mate and produce offspring. Observing the vaginal plug was considered the first day of pregnancy. On the 12.5th day of pregnancy, the VPA group received intraperitoneal (i.p.) injections of VPA (500 mg/kg/2 ml) to induce autism-like features in offspring. Instead, the SAL group received the same volume of saline (2 ml/kg).

Drugs

The pure red saffron powder was kindly supplied by the Pharmaceutical Research Center, Mashhad University of Medical Sciences, Mashhad, Iran (The specimen number of the plant is 134–0319-1) and was dissolved in saline (SAL) and injected daily (30 mg/kg/2 ml, i.p.). Crocin (17304, Sigma-Aldrich Co, USA) was dissolved in SAL and injected similarly (15 or 30 mg/kg/2 ml, i.p.). Drug doses were chosen based on our and other studies (Ghadoust article). The SAL group received the same volume of Saline daily (2 ml/kg).

Behavioral tests

Hot plate

In this test, the temperature of the plate was set at 52 °C. After placing the animal on this plate, the response to pain (licking the paws or jumping out of the plastic container on the hot plate) was recorded in seconds. We set a cutoff test time to 20 s to prevent tissue damage.

Rotarod test

The rotarod device was used to examine motor coordination and balance activities. To evaluate the balance activity, the animals were first placed on a horizontal bar to adapt. The device was set to a speed of 10 rpm to check the balance, and the time spent on the horizontal bar and balancing was recorded.

Elevated plus maze (EPM)

EPM was used to measure anxiety-like behaviors. This device was made of wood and had 4 arms shaped like a cross. The two arms of this device had no walls and dimensions of 50×10 cm. The other two arms had dark walls with dimensions of 40×50×10 cm on both sides. The height of this device from the ground was 50 cm. The animals were placed in the central area facing an open arm and moved freely in the device for 5 min. The number of entries in the open and closed arms and the total time spent in the open and closed arms were measured. Entering an open or closed arm was when all four legs of the animal were in the desired arm.

Open field (OF)

This device was an open square box with dimensions of 70×70×45 cm made of Plexiglas. A black floor formed the test environment in which a video-tracking system for 5 min monitored the behavior. The area of the field was divided into two parts, the peripheral and the central zones. The increase in time spent and the number of entries in the center was considered a decrease in anxiety-like behavior. Also, the mean speed and the total traveled distance were considered motor activities.

T-maze

T-maze was constructed of black Plexiglas, including stem and arms 35 cm long, 10 cm wide, and 15 cm high. Horizontal sliding doors separated the target arms and the start box from the central way. A lamp positioned 2 m above the stem to provide about 50-lx illumination inside the task. The animals were first submitted to an acquisition phase in which they entered twice the same arm of the maze to find a reward at the end of the target arm. During the memory test, 24 h later, animals remained for 30 s in the start box. Then, the door was opened, and animals were free to enter each goal arm within the maze, and the time spent in each arm was measured.

Measurements of oxidative stress markers in the brain

At the end of the study, all rats were sacrificed. The brains were isolated, and a fraction of that was washed in cold 0.9% saline and kept at -70°C until used to prepare homogenates with a homogenizer (Polytron PT 2100, KINEMATICA AG, Switzerland). For malondialdehyde (MDA) assays, a fraction of tissue was homogenized (1:10 w/v) in cold 1.15% KCl. Homogenates for GSH and CAT measurements were prepared in a ratio of 100 mg tissue to 1 ml phosphate buffer (50 mmol/l; pH 7.5) containing 1 mM EDTA. The supernatants obtained after centrifugation at $20,000\times g$ for 10 min at 4°C were used for biochemical analyses. The Bradford method used bovine serum albumin as a standard to determine the total protein level in supernatants.

Statistical analysis

All Data are presented as mean $\pm$ SEM. Two-way or three-way repeated-measures ANOVA was used, followed by Tukey's post hoc test. A $P < 0.05$ was considered statistically significant.

Experimental groups

On the 30th day after birth (before measuring behavioral tests and drug injections), the control male pups were randomly divided into the SH-SAL, SH-CRO15, SH-CRO30, and SH-SAF30 groups ($n=6$ in each group, SH: Sham). Similarly, autistic pups were divided into the AUT-SAL, AUT-CRO15, AUT-CRO30, and AUT-SAF30 groups. First-round behavioral tests (before treatments) were performed on days 30–33 followed by 4 weeks of drug injections as treatment, starting from day 34 until 61 (28 days), including saffron (30 mg/kg, i.p.) and crocin (15 and 30 mg/kg, i.p.) or saline (2 ml/kg, i.p.). Then, the second-round behavioral tests (after treatments) were performed again on days 62–65 to assess the effects of the treatments on behaviors and cognitive functions. In the end, the animals were sacrificed under deep anesthesia, and their brains were dissected to evaluate the oxidative stress markers, including malondialdehyde (MDA), glutathione (GSH), and catalase (CAT) using a commercial kit and the ELISA method (Fig. 1).

Results

Hot plate test results

Data from the measurement of response to pain (RP) are shown in Fig. 2A. A three-way repeated-measures ANOVA (times $\times$ groups $\times$ treatments) on RP indicated no

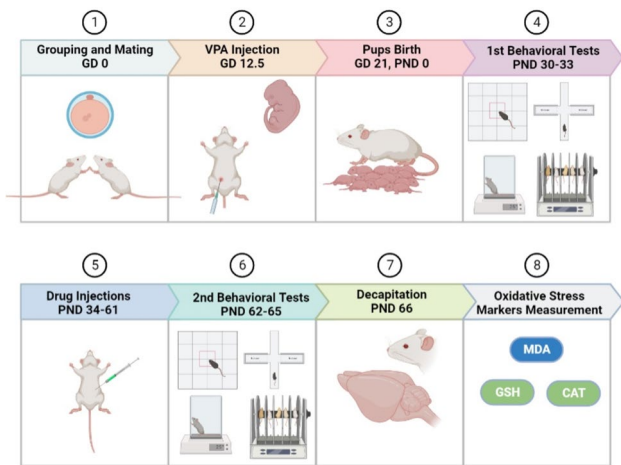


Fig. 1 Timeline of behavioral tests and drug injections (see Materials and Methods for more detail). GD: Gestation day, PND: Postnatal day

significant effect of times (time of behavioral measurements) ($F_{1,20}=2.44$, $p=0.13$), a significant effect of groups ($F_{1,20}=63.58$, $p<0.0001$), no significant effect of treatments ($F_{3,60}=1.07$, $p=0.36$), significant interactions between times and groups ($F_{1,20}=24.98$, $p<0.0001$), times and treatments ($F_{3,60}=2.95$, $p=0.03$), groups and treatments ($F_{3,60}=4.34$, $p=0.007$) and times, groups and treatments ($F_{3,60}=5.29$, $p=0.002$). Between-group comparisons indicated that the RP on day 30th after birth (before treatment) was increased in the AUT-SAL, AUT-CRO15, AUT-CRO30, and AUT-SAF30 groups than in the SH-SAL, SH-CRO15, SH-CRO30, and SH-SAF30 groups, respectively ($p<0.01$). Moreover, RP on day 62nd after birth (after treatment) was increased in the AUT-SAL group than in the SH-SAL group ($p<0.01$). RP was decreased in the AUT-CRO15, AUT-CRO30, and AUT-SAF30 groups than in the AUT-SAL group ($p<0.01$).

Briefly, VPA injection into mother rats during pregnancy induced autistic-like behaviors in their offspring. Response to pain in the autistic groups was significantly increased compared to the control groups. This finding showed that the experimental model of autism prolongs the response to peripheral thermal pain. Saffron and crocin treatment significantly ameliorated the autistic-like behaviors and response to peripheral thermal pain compared to the saline treatment.

Rotarod test results

Data from the measurement of balance times (BT) are shown in Fig. 2B. A three-way repeated-measures ANOVA (times $\times$ groups $\times$ treatments) on BT indicated significant effects of times (time of behavioral measurements) ($F_{1,20}=32.99$, $p<0.0001$), groups ($F_{1,20}=46.46$, $p<0.0001$), and treatments ($F_{3,60}=3.47$, $p=0.02$),

and significant interactions between times and groups ($F_{1,20}=13.23$, $p=0.001$), times and treatments ($F_{3,60}=3.71$, $p=0.01$), groups and treatments ($F_{3,60}=4.70$, $p=0.005$) and times, groups and treatments ($F_{3,60}=3.48$, $p=0.02$). Between-group comparisons indicated that the BT on day 30th after birth (before treatment) was decreased in the AUT-SAL, AUT-CRO15, AUT-CRO30, and AUT-SAF30 groups than in the SH-SAL, SH-CRO15, SH-CRO30, and SH-SAF30 groups, respectively ($p<0.05$). Moreover, BT on day 62nd after birth (after treatment) was decreased in the AUT-SAL group than in the SH-SAL group ($p<0.01$). BT was increased in the AUT-CRO15, AUT-CRO30, and AUT-SAF30 groups than in the AUT-SAL group ($p<0.01$).

Briefly, VPA injection into mother rats during pregnancy induced autistic-like behaviors in their offspring. Balance times in the autistic groups were significantly decreased compared to the control groups. This finding showed that the experimental model of autism decreased balance time. Saffron and crocin treatment significantly ameliorated the autistic-like behaviors and balance time compared to the saline treatment.

EPM test results

Data from the open arms time (OAT) are shown in Fig. 2C. A three-way repeated-measures ANOVA (times $\times$ groups $\times$ treatments) on OAT indicated significant effects of times (time of behavioral measurements) ($F_{1,20}=8.38$, $p=0.008$), groups ($F_{1,20}=86.11$, $p<0.0001$), and treatments ($F_{3,60}=3.06$, $p=0.03$), and a significant interaction between times and groups ($F_{1,20}=15.08$, $p=0.0009$), no interaction between times and treatments ($F_{3,60}=1.49$, $p=0.22$), and significant interactions between groups and treatments ($F_{3,60}=2.63$, $p=0.05$) and times, groups and treatments ($F_{3,60}=4.35$, $p=0.007$). Between-group comparisons indicated that the OAT on day 31st after birth (before treatment) was decreased in the AUT-SAL, AUT-CRO15, AUT-CRO30, and AUT-SAF30 groups than in the SH-SAL, SH-CRO15, SH-CRO30, and SH-SAF30 groups, respectively ($p<0.01$). Moreover, OAT on day 63rd after birth (after treatment) was decreased in the AUT-SAL group than in the SH-SAL group ($p<0.01$). OAT was increased in the AUT-CRO15, AUT-CRO30, and AUT-SAF30 groups than in the AUT-SAL group ($p<0.01$).

Data from the open arms entry (OAE) are shown in Fig. 2D. A three-way repeated-measures ANOVA (times $\times$ groups $\times$ treatments) on OAE indicated significant effects of times (time of behavioral measurements) ($F_{1,20}=5.33$, $p=0.03$), and groups ($F_{1,20}=22.77$, $p=0.0001$), no significant effect of treatments ($F_{3,60}=1.24$, $p=0.3$), a significant interaction between times and groups ($F_{1,20}=12.53$, $p=0.002$), but no significant interactions between times and treatments ($F_{3,60}=0.37$, $p=0.77$), groups

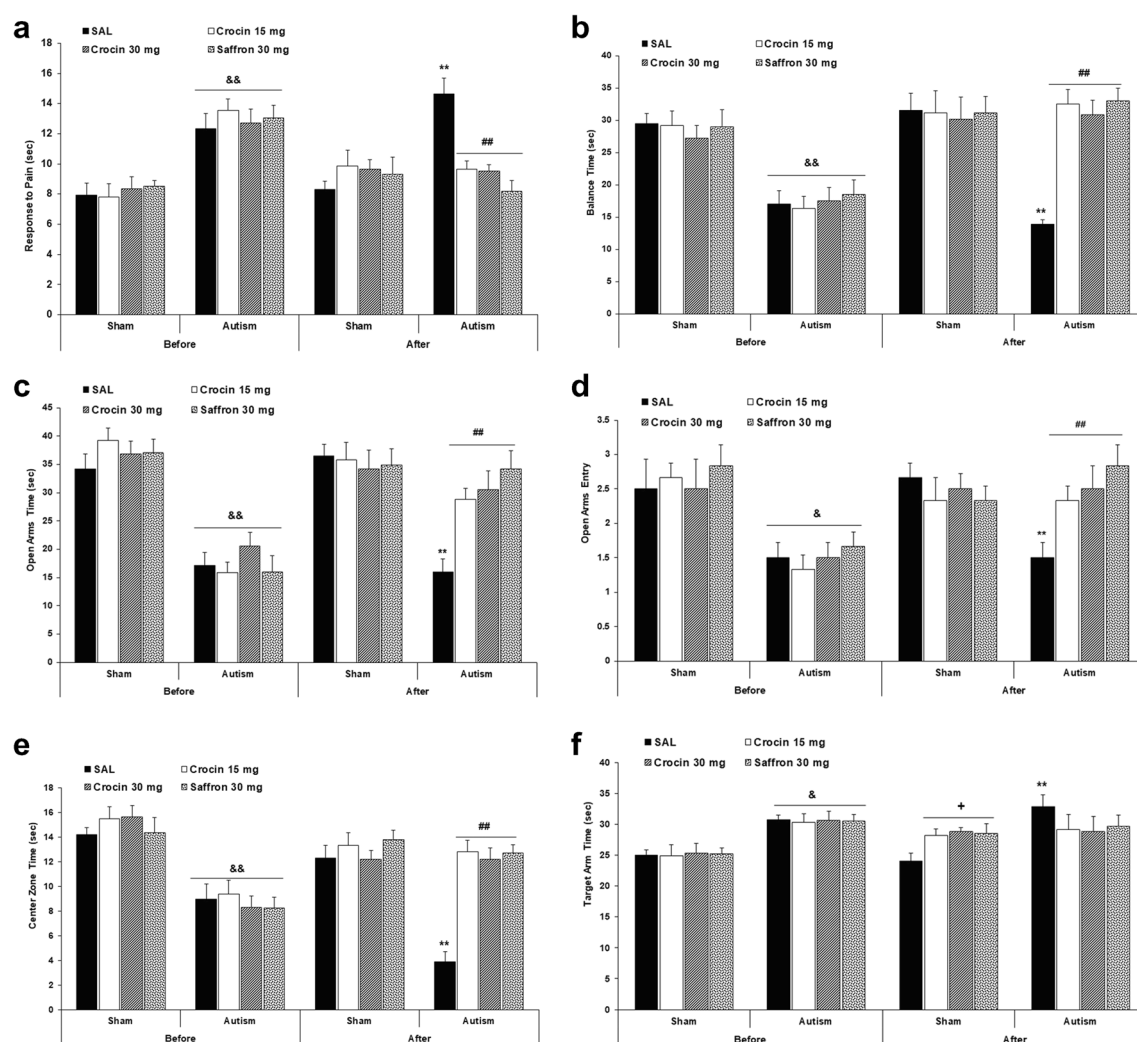


Fig. 2 The effects of saffron and crocin on VPA-induced behavioral, motor, and cognitive features in male offspring rats. **A:** Hot plate test to evaluate pain response. **B:** Rotarod test to evaluate balance times. **C and D:** Elevated plus maze to evaluate anxiety-like behaviors. **E:** Open field to evaluate anxiety-like behaviors. **F:** T-maze to evaluate

spatial memory. Data are expressed as mean $\pm$ SEM, $n=6$. * $P<0.05$ and ** $P<0.01$ compared to the SH-SAL group (healthy). # $P<0.05$ and ## $P<0.01$ compared to the AUT-SAL group (autistic). SH: Sham, SAL: Saline, AUT: Autism

and treatments ($F_{3,60}=1.29$, $p=0.28$) and times, groups and treatments ($F_{3,60}=1.99$, $p=0.12$). Between-group comparisons indicated that the OAE on day 31st after birth (before treatment) was decreased in the AUT-SAL, AUT-CRO15, AUT-CRO30, and AUT-SAF30 groups than in the SH-SAL, SH-CRO15, SH-CRO30, and SH-SAF30 groups, respectively ($p<0.01$). Moreover, OAT on day 63rd after birth (after treatment) was decreased in the AUT-SAL group than in the SH-SAL group ($p<0.01$). But OAE was increased in the AUT-CRO15, AUT-CRO30, and AUT-SAF30 groups than in the AUT-SAL group ($p<0.01$).

Briefly, VPA injection into mother rats during pregnancy induced autistic-like behaviors in their offspring. Time spent in open arms and open arm entries in the autistic groups was significantly decreased compared to the control groups.

These findings showed that the experimental model of autism increased anxiety-like behaviors. Saffron and crocin treatment significantly ameliorated the autistic and anxiety-like behaviors compared to the saline treatment.

OF test results

Data from the center zone times (CZT) are shown in Fig. 2E. A three-way repeated-measures ANOVA (times $\times$ groups $\times$ treatments) on CZT indicated no significant effect of times (time of behavioral measurements) ($F_{1,20}=0.14$, $p=0.9$), but significant effects of groups ($F_{1,20}=125.69$, $p<0.0001$), and treatments ($F_{3,60}=7.71$, $p=0.0002$), a significant interaction between times and groups ($F_{1,20}=21.41$, $p=0.0002$), times and treatments

($F_{3, 60} = 4.54$, $p = 0.006$), no significant interaction between groups and treatments ($F_{3, 60} = 2.05$, $p = 0.1$) and a significant interaction between times, groups and treatments ($F_{3, 60} = 6.48$, $p = 0.0007$). Between-group comparisons indicated that the CZT on day 32nd after birth (before treatment) was decreased in the AUT-SAL, AUT-CRO15, AUT-CRO30, and the AUT-SAF30 groups than in the SH-SAL, SH-CRO15, SH-CRO30, and the SH-SAF30 groups, respectively ($p < 0.05$). Moreover, CZT on day 64th after birth (after treatment) was decreased in the AUT-SAL group than in the SH-SAL group ($p < 0.01$). OAT was increased in the AUT-CRO15, AUT-CRO30, and AUT-SAF30 groups than in the AUT-SAL group ($p < 0.01$).

Briefly, VPA injection into mother rats during pregnancy induced autistic-like behaviors in their offspring. The time spent in the central zone of the open field in the autistic groups was significantly decreased compared to the control groups. This finding showed that the experimental model of autism increased anxiety-like behaviors. Saffron and crocin treatment significantly ameliorated the autistic and anxiety-like behaviors compared to the saline treatment.

T-maze test results

Data from the measurement of target arm times (TAT) are shown in Fig. 2F. A three-way repeated-measures ANOVA (times $\times$ groups $\times$ treatments) on TAT indicated no significant effect of times (time of behavioral measurements) ($F_{1, 20} = 1.68$, $p = 0.20$), a significant effect of groups ($F_{1, 20} = 29.05$, $p < 0.0001$), no significant effect of treatments ($F_{3, 60} = 0.19$, $p = 0.99$), no significant interactions between times and groups ($F_{1, 20} = 0.90$, $p = 0.35$), times and treatments ($F_{3, 60} = 0.02$, $p = 0.99$), groups and treatments ($F_{3, 60} = 0.51$, $p = 0.67$) and times, groups and treatments ($F_{3, 60} = 0.33$, $p = 0.80$). Between-group comparisons indicated that the TAT on day 33rd after birth (before treatment) was increased in the AUT-SAL, AUT-CRO15, AUT-CRO30, and AUT-SAF30 groups than in the SH-SAL, SH-CRO15, SH-CRO30, and SH-SAF30 groups, respectively ($p < 0.05$). Moreover, TAT on day 65th after birth (after treatment) was increased in the SH-CRO15, SH-CRO30, and SH-SAF30 groups than in the SH-SAL group ($p < 0.05$) and increased in the AUT-SAL than in the SH-SAL ($p < 0.01$).

Briefly, target arm times in the autistic groups were significantly increased compared to the control groups. This finding showed that the experimental model of autism did not negatively affects cognitive functions and increased spatial memory. Saffron and crocin treatment significantly increased spatial memory in healthy subjects.

Brain oxidative stress parameters

Data from the brain's oxidative stress parameters are shown in Fig. 3A–C.

A two-way ANOVA (groups $\times$ treatments) on MDA levels showed significant effects of groups ($F_{1, 16} = 191.41$, $P < 0.0001$) and treatments ($F_{3, 16} = 22.69$, $P < 0.0001$) and significant interaction between groups and treatments ($F_{3, 16} = 14.73$, $P < 0.0001$). Between-group comparisons indicated that the MDA level on day 66th after birth (after treatment) was increased in the AUT-SAL than in the SH-SAL ($p < 0.01$). Moreover, MDA levels were decreased in the AUT-CRO15, AUT-CRO30, and AUT-SAF30 than in the AUT-SAL ($p < 0.01$) (Fig. 3A).

A two-way ANOVA (groups $\times$ treatments) on GSH levels showed significant effects of groups ($F_{1, 16} = 58.82$, $P < 0.0001$) and treatments ($F_{3, 16} = 11.95$, $P = 0.0002$) but no significant interaction between groups and treatments ($F_{3, 16} = 2.45$, $P = 0.1$). Between-group comparisons indicated that the GSH level on day 66th after birth (after treatment) was decreased in the AUT-SAL than in the SH-SAL ($p < 0.01$). Moreover, GSH levels were increased in the AUT-CRO15 ($p < 0.05$), AUT-CRO30 ($p < 0.01$), and AUT-SAF30 ($p < 0.01$) than in the AUT-SAL (Fig. 3B).

A two-way ANOVA (groups $\times$ treatments) on CAT levels showed significant effects of groups ($F_{1, 16} = 80.17$, $P < 0.0001$) and treatments ($F_{3, 16} = 7.86$, $P = 0.001$) but no significant interaction between groups and treatments ($F_{3, 16} = 2.17$, $P = 0.13$). Between-group comparisons indicated that the CAT level on day 66th after birth (after treatment) was decreased in the AUT-SAL than in the SH-SAL ($p < 0.01$). Moreover, CAT levels were increased in the AUT-CRO15, AUT-CRO30, and AUT-SAF30 than in the AUT-SAL ($p < 0.01$) (Fig. 3B).

Briefly, VPA injection into mother rats during pregnancy induced autistic-like features in their offspring. The brain levels of MDA were significantly increased, but the levels of GSH and CAT were significantly decreased in the autistic groups compared with the SAL (healthy) groups. These findings showed that the experimental model of autism causes oxidative stress. Saffron and crocin treatment significantly decreased MDA and increased GSH and CAT levels compared to the saline treatment.

Discussion

The main finding of this study includes 1. VPA injection into female rats during pregnancy induced autistic-like behaviors in their male offspring. 2. Saffron extract and its active constituent, crocin, improved autistic-like symptoms during adolescence and puberty. 3. GSH and CAT levels were decreased, and the MDA level was increased in the

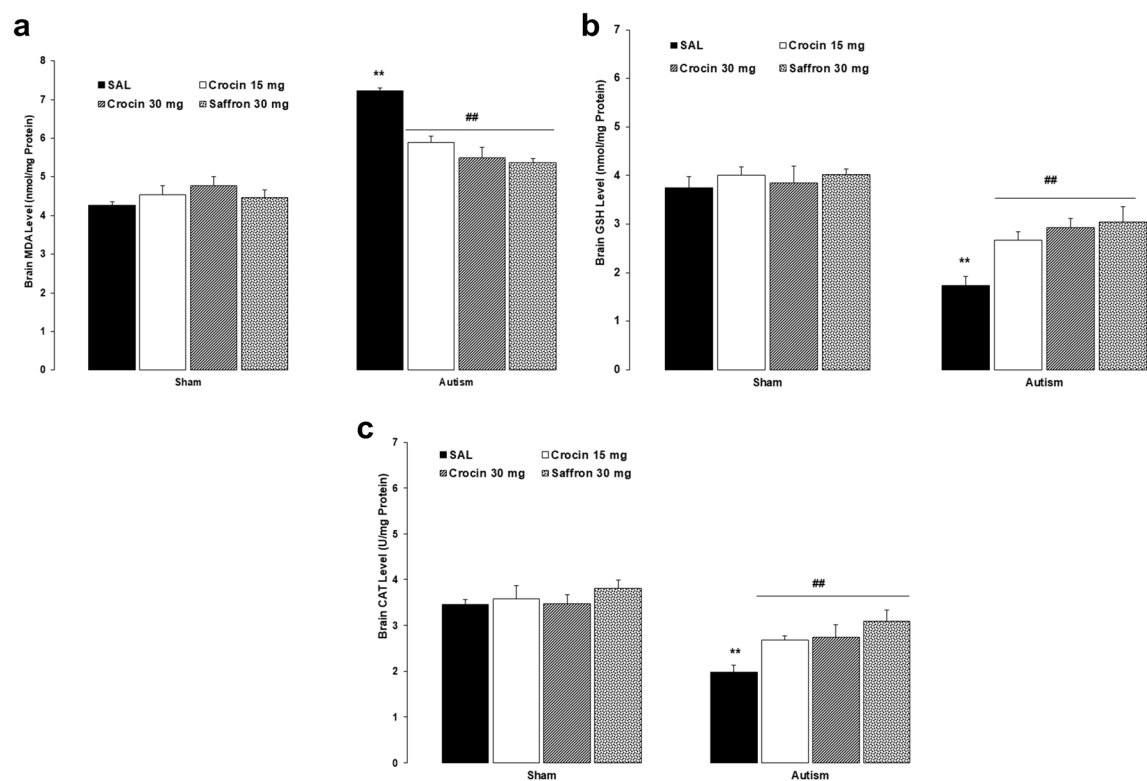


Fig. 3 The effects of saffron and crocin on VPA-induced brain oxidative stress markers in offspring rats. **A:** Malondialdehyde. **B:** Glutathione. **C:** Catalase. Data are expressed as mean $\pm$ SEM, $n=6$.
 * $P<0.05$ and ** $P<0.01$ compared to the SH-SAL group (healthy). # $P<0.05$ and ## $P<0.01$ compared to the AUT-SAL group (autistic). SH: Sham, SAL: Saline, AUT: Autism

brain of autism-induced offspring rats. 4. Saffron and crocin improved the brain’s oxidative stress parameters (Table 1).

Our findings show that the VPA-induced animal model of autism enhanced pain response and anxiety-like behaviors, impaired motor functions, and reduced balance power in rats. 28 days of treatment with saffron and crocin significantly improved behavioral outcomes compared to the saline-treated groups. Saffron and crocin increased spatial memory in healthy subjects compared to the saline-treated control group. Interestingly, the results of the T-maze task

revealed that not only was spatial memory not impaired in VPA-exposure rats, but also significantly enhanced. Such contradictory results are not rare. Several studies have reported that spatial memory is increased following animal models of autism (Abbarin et al. 2023; Ghahremani et al. 2022). Even such discrepancy has been reported for the other behavioral tasks. For example, Mychasiuk et al., using an EPM task, have shown that VPA-induced autistic offspring rats spent a greater time in open arms compared to the control groups (Mychasiuk et al. 2012). Regarding the T-maze

Table 1 Summary of behavioral tests and brain oxidative stress markers assessed in VPA-induced autistic rats before and after 28 days of treatment with saffron and crocin

Behavioral Tests	Results Before Treatment With Saffron and Crocin	Results After Treatment With Saffron and Crocin
Hot Plate	↑ Response to Pain	↓ Response to Pain
Rotarod Test	↓ Balance Time	↑ Balance Time
Elevated Plus Maze	↓ Open Arms Time ↓ Open Arms Entry	↑ Open Arms Time ↑ Open Arms Entry
Open Field	↓ Center Zone Time	↑ Center Zone Time
T-Maze	↑ Target Arm Time	↑ Target Arm Time
Oxidative Stress Markers	Saline-treated Animals	Saffron and Crocin-treated Animals
MDA	↑	↓
GSH	↓	↑
CAT	↓	↑

task, usually, intact animals prefer to explore the novel arm instead of the first one. An explanation for the above finding could be that autistic rats may have a greater tendency to repeat a behavior, as previously reported by Markram et al. (Markram et al. 2008). Using a Y-maze task, they show that VPA-exposed rats have increased repetitive behavior and tend to re-enter the same previously entered side of the Y-maze, which aligns with the present study.

Reportedly, the brains of rats receiving VPA undergo anatomical abnormalities in the cerebellum, amygdala, hippocampus, and frontal and parietal lobes, similar to those in children with autism (Schneider and Przewłocki 2005a). Therefore, the VPA-induced autism model has both structural and behavioral characteristics identical to autistic patients and so can be used as an appropriate model for evaluating autistic behaviors (Schneider and Przewłocki 2005a). Our results showed that this experimental model of autism leads to impaired balance and movement, consistent with previous findings (Dufour-Rainfray et al. 2010; Edalatmanesh and Samimi 2018). Edalatmanesh et al. showed that behavioral and balance disturbances in VPA-induced autistic rats compared to the control group might be due to decreased BDNF levels in some brain areas, such as the cerebellum or damage to Purkinje cells in the cerebellum (Edalatmanesh and Samimi 2018). We showed that 4 weeks of treatment with saffron and its main ingredient, crocin, significantly improved behavioral disorders in rats. Previous studies have shown that crocin increases the level of BDNF in the brain, which may be one of the mechanisms by which saffron and crocin improve autistic behaviors (Ghalandari-Shamami et al. 2021; Sun et al. 2020). Moreover, autism changes the levels of neurotransmitters in the brain, followed by a decrease in serotonin, especially in the hippocampus (Zieminska et al. 2021). It has been suggested that behavioral and social disorders in autism may cause by changes in the activity of the serotonergic system (Dufour-Rainfray et al. 2010). Saffron and crocin modulate dopamine, noradrenaline, and serotonin neurotransmitter systems, activate the MKP-1/CREB/BDNF pathway, and establish a balance between proBDNF and mBDNF, thereby regulating neurotransmitter levels and improve behavioral disorders (Wang et al. 2019).

Reactive oxygen species (ROS) play an essential role in the pathophysiology of neurological diseases. Increased nitric oxide and lipid peroxidation levels in autism, followed by a change in the level of oxidative stress parameters, may be one of the leading causes of symptoms in this disease (Söğüt et al. 2003; Yarıbeygi et al. 2018c). Brain tissue is vulnerable to oxidative damage (Metodiewa and Koška 2000). Our findings showed increased MDA levels and decreased GSH and CAT levels, indicating a weakening of the antioxidative system and oxidative stress induction in brain tissue. Nevertheless, saffron and crocin significantly reversed these changes, ameliorating oxidative stress in rats'

brains. Our previous study has shown that saffron extract and crocin can modulate neuropathic pain (Safakhah et al. 2016) and improve spatial memory, learning disorders, and oxidative stress following chronic stress (Ghadroost et al. 2011). The antioxidant effect of crocin has been shown to improve oxidative stress by increasing the activity of antioxidant enzymes such as glutathione peroxidase, glutathione reductase, and superoxide dismutase (Ghadroost et al. 2011; Yarıbeygi et al. 2018a, b, c; Yarıbeygi et al. 2019). Vakili et al. showed that crocin prevents the increase in MDA levels and increases the superoxide dismutase (SOD) and glutathione peroxidase (GPX) enzymes' activities (Vakili et al. 2011). Considering that oxidative stress is a main underlying factor in the pathogenesis of autism, the beneficial effects of saffron and crocin on the animal's behaviors in the present study may be through modifying the oxidative stress.

Previous studies have shown an increase in the markers of oxidative stress in autism, and these factors, like other environmental and genetic agents, are involved in the pathophysiology of this disorder (Bjørklund et al. 2020). Increasing pre-oxidative factors, including NO, xanthin oxidase, and homocysteine, and decreasing antioxidative elements, such as ceruloplasmin, transferrin, SOD, GPX, catalase, and glutathione, can cause oxidative stress (Bjørklund et al. 2020). Oxidative stress leads to cell membrane lipids defects, mitochondrial dysfunction, inflammation, immunodeficiency in children, and autistic-like behaviors (Chen et al. 2021). Also, clinical research has shown an improvement in the individual behaviors of children with autism treated with antioxidants (Chauhan and Chauhan 2006). VPA reduces GSH and CAT enzymes' activities and increases MDA levels, but antioxidants of piperine improve these factors and reduce MDA levels (Pragnya et al. 2014). GSH and CAT are two antioxidant enzymes involved in the antioxidative system, so reducing their activity may be translated into reducing the antioxidant capacity (Chen et al. 2021). A decrease in GSH and SOD activities has been correlated with a decrease in neuroprotection against oxidative stress, suggesting the importance of the antioxidative stress system in normalizing these changes and autism treatment (Ghanizadeh et al. 2012).

This study has two limitations. Firstly, while we recorded and reported the animals' initial weight, we could not obtain their weights during the experiments. Secondly, autism spectrum disorders (ASD) in animal models are described by core ASD-like behaviors (social communication deficits and repetitive behaviors) (Kim et al. 2013a, b) and some comorbid traits, including anxiety-like behavior, motor deficits, and abnormal sensory processing (Argyropoulos et al. 2013). So far, despite the prevalence of comorbidities in autism patients, animal models have focused mainly on characterizing core behavioral traits (Argyropoulos et al. 2013). Maternal exposure to VPA not only produces deficits in (Patterson 2011a, b) core ASD-like behaviors (social

interaction and repetitive-stereotyped behaviors) (Markram et al. 2008; Schneider and Przewłocki 2005b) but also leads to one or more comorbid features, including altered sensory processing, anxiety-like behavior, and disturbed motor functions (Argyropoulos et al. 2013). Accordingly, mainly due to this reason, we measured some comorbid traits in the present VPA-induced animal model of ASD. Additionally, due to some technical issues, we did not measure core ASD-like behaviors, which the readers should notice as the limitation of this work.

Conclusion

In summary, the present study showed that intraperitoneally injection of VPA into female rats on day 12.5 of pregnancy induces autistic-like behaviors in their male offspring. Brain oxidative stress is involved in the pathophysiology of autistic-like behaviors. 28 days of treatment with saffron and crocin improved the brain's oxidative stress parameters and ameliorated autistic-like behaviors such as increased pain responses, anxiety-like behaviors, and motor and balance deficits. Practically, our findings from this pre-clinical animal study may highlight the beneficial effects of saffron and crocin, as herbal-based pharmacologic agents promising drugs, in treating autistic patients. However, further histological evidence is needed to confirm their efficacy.

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Authors' contributions A.A.V. and A.R.P. designed the experiment. S.A.S., P.T., D.A., K.S., H.Y., P.R.A., and A.A.V. conducted the research, collected data, and carried out the lab work. A.A.V., H.Y., and A.R.P. carried out the statistical analysis and mostly drafted the manuscript. A.A.V., A.R.P., and P.R.A. coordinated and supervised the study. All authors approved the manuscript.

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Data availability The datasets generated and analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

All the authors have approved this paper's contents and agreed to the "Metabolic Brain Disease" journal publication policies.

Ethical considerations The Ethical Review Board of Semnan University of Medical Sciences, Semnan, Iran approved the experimental protocol. All experiments were completed in agreement with the National Institutes of Health Guide for the Care and Use of Laboratory Animals. (NO: IR.SEMUMS.REC.1397.144).

Conflict of interest The authors declare no conflict of interest.

Consent to participate This research involved non-human subjects (Rats).

Consent to publish This research involved non-human subjects (Rats).

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